

Implementation of Newton Raphson Method in C, using $f(x) = x^3 - 2x - 5$ and $f(x) = \sin x + x^2 - 1$

Code:

```
#include<stdio.h>
#include<math.h>

double f(double x){
    return (x*x*x)-2*x-5;
}

double fdx(double x){
    return (3*x*x)-2;
}

int main(){
    double x0,x1,e;
    int i=0,n;

    printf("Enter initial guess: ");
    scanf("%lf",&x0);
    printf("Enter error: ");
    scanf("%lf",&e);
    printf("Enter maximum iteration: ");
    scanf("%d",&n);

    do{
        if(fdx(x0)==0){
            printf("Division by zero error");
            return 0;
        }

        x1=x0-(f(x0)/fdx(x0));
        printf("Iteration number: %d x0=%.10f\n",i,x1);

        i++;
        if(fabs(x1-x0)<=e){
            printf("Final root: %lf",x1);
            break;
        }
    }
```

```
        x0=x1;

    }while(i<=n);

    return 0;
}
```

Output:

For equation: $f(x) = x^3 - 2x - 5$

Enter initial guess: 2

Enter error: 0.000000001

Enter maximum iteration: 5

Iteration number: 0 x0=2.1000000000

Iteration number: 1 x0=2.0945681211

Iteration number: 2 x0=2.0945514817

Iteration number: 3 x0=2.0945514815

Final root: 2.094551

For equation: $f(x) = \sin x + x^2 - 1$

Enter initial guess: 1

Enter error: 0.000000001

Enter maximum iteration: 5

Iteration number: 0 x0=0.6687516352

Iteration number: 1 x0=0.6370680331

Iteration number: 2 x0=0.6367326888

Iteration number: 3 x0=0.6367326508

Iteration number: 4 x0=0.6367326508

Final root: 0.636733